

13) $\log_4 6 \approx 1.3$
 $\log_4 7 \approx 1.4$
 $\log_4 10 \approx 1.7$
 Find $\log_4 \frac{7}{16}$

14) $\log_5 9 \approx 1.4$
 $\log_5 7 \approx 1.2$
 $\log_5 6 \approx 1.1$
 Find $\log_5 150$

15) $\log_3 10 \approx 2.1$
 $\log_3 8 \approx 1.9$
 $\log_3 11 \approx 2.2$
 Find $\log_3 \frac{11}{30}$

16) $\log_6 4 \approx 0.8$
 $\log_6 10 \approx 1.3$
 $\log_6 9 \approx 1.2$
 Find $\log_6 \frac{9}{40}$

Use the properties of logarithms and the logarithms provided to rewrite each logarithm in terms of the variables given.

17) $\log_6 4 = R$
 $\log_6 11 = S$
 $\log_6 10 = T$
 Find $\log_6 150$

18) $\log_8 6 = X$
 $\log_8 11 = Y$
 $\log_8 9 = Z$
 Find $\log_8 \frac{512}{11}$

19) $\log_4 6 = A$
 $\log_4 7 = B$
 $\log_4 9 = C$
 Find $\log_4 \frac{8}{21}$

20) $\log_6 11 = X$
 $\log_6 10 = Y$
 $\log_6 4 = Z$
 Find $\log_6 12100$

21) $\log_9 7 = X$
 $\log_9 6 = Y$
 $\log_9 8 = Z$
 Find $\log_9 \frac{9}{448}$

22) $\log_9 10 = X$
 $\log_9 12 = Y$
 $\log_9 7 = Z$
 Find $\log_9 \frac{5}{864}$